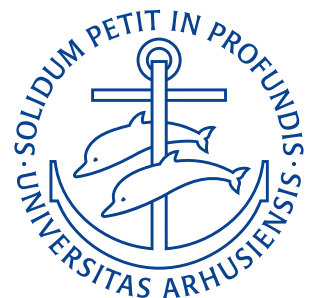


Department of Mechanical and Production Engineering
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Assignment 3: The Elastic Problem

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Definition of Problem 1

Consider the bi-layer prismatic bar represented in Figure 3, under tension forces N applied at both ends. Let us assume that the two layers, made of different materials, remain perfectly attached to each other.

Specifically, we assume both materials to be isotropic linearly elastic, i.e., governed by Hooke's law. We further assume the red material (the top layer) to have a Young modulus E_1 and a Poisson ratio ν , and the green material (the bottom layer) to have a Young modulus E_2 and a Poisson ratio ν .

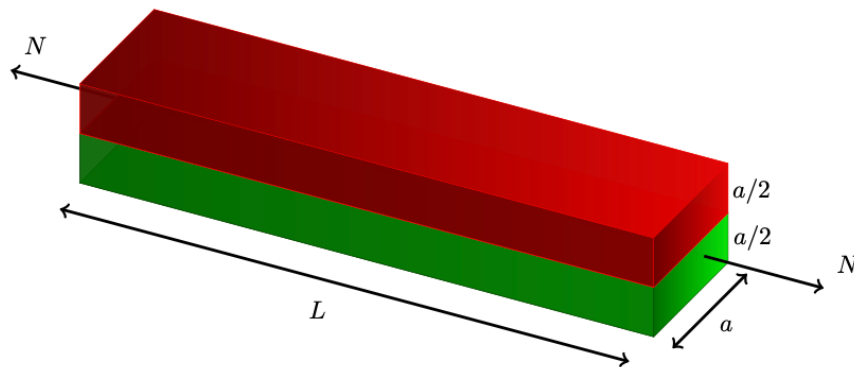


Figure 3: Bilayer bar under pulling force N .

Problem 1.1

Applying the De Saint Venant principle, determine the states of stress and strain within the continuum. Pay special attention to what happens as the interface between the two materials. Hint: The displacement field should be continuous across the interface while the strain and stress fields need not be.

Assumptions

- Any extension of the bar is purely axial, i.e. the bar does not bend

Derivation

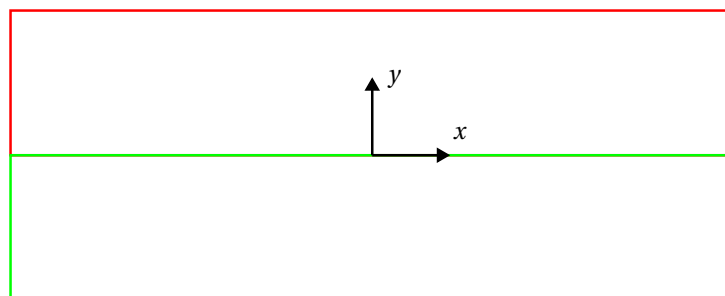


Figure 3.1: Coordinate system for Problem 1.

We employ the coordinate axes shown on Figure 3.1, where we let x be the long direction of the bar and y be the vertical, with the material interface at $y = 0$. We denote by material 1 the red/top layer and by material 2 the

green/bottom layer. Each layer has area

$$A_1 = A_2 = \frac{a^2}{2}. \quad (1)$$

Saint-Venant's principle lets us ignore the local loading effects, far away from the loaded ends, where we can treat the bar as being in a uniform axial extension state.

Because the two materials are perfectly attached at the interface, they cannot extend by different amount in the x -direction. Hence, the axial strain must be the same in both layers

$$\epsilon_{xx}^{(1)} = \epsilon_{xx}^{(2)} = \epsilon_0.$$

So, the displacement field can be written as

$$u_x = \epsilon_0 x \quad (2)$$

for both materials. Since the Poisson ratio is the same in both layers, the lateral contraction will also be compatible:

$$u_y = -\nu \epsilon_0 y, \quad u_z = -\nu \epsilon_0 z. \quad (3)$$

So the displacement field is therefore continuous across the interface.

For uniaxial stress, Hooke's law gives

$$\sigma_{xx}^{(i)} = E_i \epsilon_0$$

so

$$\sigma_{xx}^{(1)} = E_1 \epsilon_0, \quad \sigma_{xx}^{(2)} = E_2 \epsilon_0. \quad (4)$$

Also, the total axial force must equal N

$$N = \sigma_{xx}^{(1)} A_1 + \sigma_{xx}^{(2)} A_2. \quad (5)$$

Substituting [Equations \(1\) and \(4\)](#) into [Equation \(5\)](#) yields:

$$N = E_1 \epsilon_0 \frac{a^2}{2} + E_2 \epsilon_0 \frac{a^2}{2}. \quad (6)$$

Now, solving [Equation \(6\)](#) for ϵ_0 gives

$$\epsilon_0 = \frac{2N}{a^2 (E_1 + E_2)}. \quad (7)$$

Substituting [Equation \(7\)](#) into [Equation \(4\)](#) gives the axial stresses in each material:

$$\sigma_{xx}^{(1)} = \frac{2NE_1}{a^2 (E_1 + E_2)} \quad (8)$$

$$\sigma_{xx}^{(2)} = \frac{2NE_2}{a^2 (E_1 + E_2)}. \quad (9)$$

I.e each material will carry a stress proportional to its Young modulus, meaning the stiffer material will carry the most stress.

From [Equations \(2\) and \(3\)](#) the strain tensor is the same for both materials:

$$\epsilon^{(1)} = \epsilon^{(2)} = \begin{bmatrix} \epsilon_0 & 0 & 0 \\ 0 & -\nu \epsilon_0 & 0 \\ 0 & 0 & -\nu \epsilon_0 \end{bmatrix}$$

where is as defined in [Equation \(7\)](#). Explicitly,

$$\epsilon = \begin{bmatrix} \frac{2N}{a^2 (E_1 + E_2)} & 0 & 0 \\ 0 & -\nu \frac{2N}{a^2 (E_1 + E_2)} & 0 \\ 0 & 0 & -\nu \frac{2N}{a^2 (E_1 + E_2)} \end{bmatrix}.$$

As only the axial stress N is applied, the stress state is described completely for each layer by $\sigma_{xx}^{(1)}$ and $\sigma_{xx}^{(2)}$ as given in Equations (8) and (9),

$$\sigma^{(1)} = \begin{bmatrix} \frac{2NE_1}{a^2(E_1+E_2)} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sigma^{(2)} = \begin{bmatrix} \frac{2NE_2}{a^2(E_1+E_2)} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

This means, that for any $E_1 \neq E_2$, the axial stress jumps at the interface as $\sigma_{xx}^{(1)} \neq \sigma_{xx}^{(2)}$.

At the interface, the displacement must be continuous, $u^{(1)} = u^{(2)}$, however the stress field does not need to be continuous here.

$$\sigma^{(1)} = \begin{bmatrix} \frac{2NE_1}{a^2(E_1+E_2)} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sigma^{(2)} = \begin{bmatrix} \frac{2NE_2}{a^2(E_1+E_2)} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\epsilon^{(1)} = \epsilon^{(2)} = \begin{bmatrix} \frac{2N}{a^2(E_1+E_2)} & 0 & 0 \\ 0 & -\nu \frac{2N}{a^2(E_1+E_2)} & 0 \\ 0 & 0 & -\nu \frac{2N}{a^2(E_1+E_2)} \end{bmatrix}$$

RESULT

Problem 1.2

What is the relationship between the axial load N and the stretch (relative change of length) $\Delta L/L$?

Derivation

The stretch in this context is just the axial strain

$$\frac{\Delta L}{L} = \epsilon_{xx} = \epsilon_0.$$

As mentioned in Problem 1.1, the two layers are perfectly bonded and must therefore have the same axial stretch

$$\epsilon_{xx}^{(1)} = \epsilon_{xx}^{(2)} = \frac{\Delta L}{L} = \epsilon_0.$$

The size of this quantity ϵ_0 was found in Problem 1.1 to be:

$$\epsilon_0 = \frac{\Delta L}{L} = \frac{2N}{a^2 (E_1 + E_2)}.$$

$\frac{\Delta L}{L} = \epsilon_0 = \frac{2N}{a^2 (E_1 + E_2)}$
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RESULT

Problem 1.3

What happens if $E_1/E_2 \rightarrow \infty$?

Derivation

From Problem 1.2 the extension was shown to be

$$\frac{\Delta L}{L} = \frac{2N}{a^2 (E_1 + E_2)}. \quad (10)$$

For $E_1/E_2 \rightarrow \infty$, Equation (10) goes toward

$$\frac{\Delta L}{L} \approx \frac{2N}{a^2 E_1}.$$

I.e. the bar behaves almost as if only the top layer exists with a cross-sectional area $a^2/2$. In this case the stresses become:

$$\begin{aligned} \sigma_{xx}^{(1)} &= \frac{2NE_1}{a^2 (E_1 + E_2)} \rightarrow \frac{2N}{a^2} \\ \sigma_{xx}^{(2)} &= \frac{2NE_2}{a^2 (E_1 + E_2)} \rightarrow 0. \end{aligned}$$

So, the top layer is load-bearing, while the bottom layer is negligible.

Further, the ratio E_1/E_2 may go to infinity in two subtly distinct ways. First, $E_1 \rightarrow \infty$ with $E_2 \neq 0$ fixed. In this case, the ratio $\Delta L/L \rightarrow 0$, meaning the whole bar essentially becomes rigid. If instead, $E_2 \rightarrow 0$, while $E_1 \not\rightarrow \infty$ (where $\not\rightarrow$ signifies a quantity *not* approaching some value), then the stretch goes to

$$\frac{\Delta L}{L} = \frac{2N}{a^2 E_1}$$

which is just the stretch of the top layer alone.

For both cases, the contribution of the bottom layer becomes negligible compared to that of the top.

$$\boxed{\frac{\Delta L}{L} \approx \frac{2N}{a^2 E_1}}$$

RESULT

Definition of Problem 2

Consider the prismatic bar represented in Figure 4, where now the two materials are arranged differently, under tension forces N applied at both ends. Let us assume that the two layers, made of different materials, remain perfectly attached to each other.

Specifically, the red material (the top layer), is characterized by E_1 and ν . The green material (the bottom layer) is characterized by E_2 and ν .

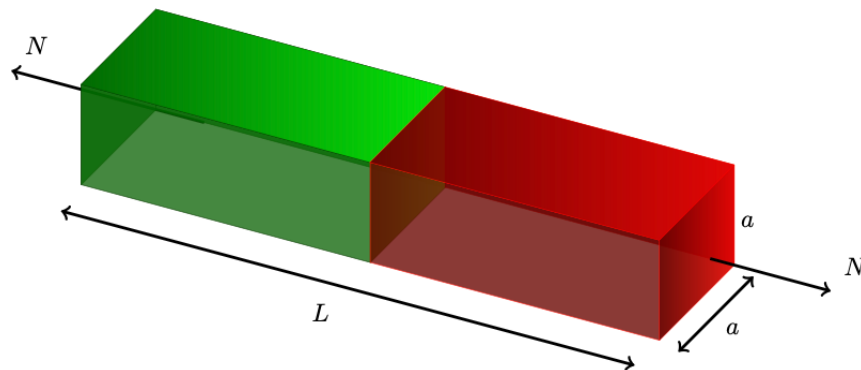


Figure 4: Composite bar under pulling force N .

Problem 2.1

Applying the De Saint Venant principle, determine the state of stress and strain within the continuum.

Derivation

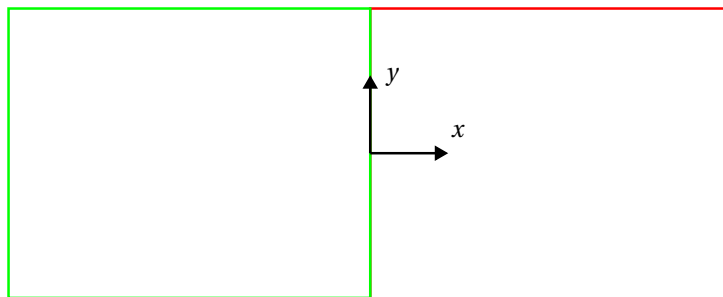


Figure 4.1: Coordinate system for Problem 2.

In this case, we employ the coordinate axes shown on Figure 4.1, where the axial direction is x . The full cross-section is

$$A = a^2.$$

Because the two materials are in series, the same axial force N must pass through both materials. Therefore, the axial stress will be the same in both materials:

$$\sigma_{xx}^{(1)} = \sigma_{xx}^{(2)} = \frac{N}{a^2}. \quad (11)$$

As N is the only applied, Equation (11) completely describes the stress state. The stress tensor away from the ends (and away from the interface between the materials) is therefore simply

$$\boldsymbol{\sigma}^{(1)} = \boldsymbol{\sigma}^{(2)} = \begin{bmatrix} \frac{N}{a^2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

This means the stress, this time, is continuous across the interface opposed to Problem 1.

Hooke's law gives

$$\epsilon_{xx}^{(i)} = \frac{\sigma_{xx}}{E_i}.$$

So

$$\epsilon_{xx}^{(1)} = \frac{N}{a^2 E_1} \quad (12)$$

$$\epsilon_{xx}^{(2)} = \frac{N}{a^2 E_2}. \quad (13)$$

Hence, for any $E_1 \neq E_2$, $\epsilon_{xx}^{(1)} \neq \epsilon_{xx}^{(2)}$. In this case the elongation of the softer material is not constrained by the stronger one, and the softer material will therefore stretch more, as the applied stress is equal. Including the Poisson contraction, the strain tensors become

$$\begin{aligned} \boldsymbol{\epsilon}^{(1)} &= \begin{bmatrix} \frac{N}{a^2 E_1} & 0 & 0 \\ 0 & -\nu \frac{N}{a^2 E_1} & 0 \\ 0 & 0 & -\nu \frac{N}{a^2 E_1} \end{bmatrix} \\ \boldsymbol{\epsilon}^{(2)} &= \begin{bmatrix} \frac{N}{a^2 E_2} & 0 & 0 \\ 0 & -\nu \frac{N}{a^2 E_2} & 0 \\ 0 & 0 & -\nu \frac{N}{a^2 E_2} \end{bmatrix}. \end{aligned}$$

Therefore, the strain will be discontinuous at the interface in this case, whilst the stress will be continuous opposed to Problem 1.

$$\begin{aligned} \boldsymbol{\sigma}^{(1)} = \boldsymbol{\sigma}^{(2)} &= \begin{bmatrix} \frac{N}{a^2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \boldsymbol{\epsilon}^{(2)} &= \begin{bmatrix} \frac{N}{a^2 E_1} & 0 & 0 \\ 0 & -\nu \frac{N}{a^2 E_1} & 0 \\ 0 & 0 & -\nu \frac{N}{a^2 E_1} \end{bmatrix} \\ \boldsymbol{\epsilon}^{(3)} &= \begin{bmatrix} \frac{N}{a^2 E_2} & 0 & 0 \\ 0 & -\nu \frac{N}{a^2 E_2} & 0 \\ 0 & 0 & -\nu \frac{N}{a^2 E_2} \end{bmatrix}. \end{aligned}$$

RESULT

Problem 2.2

What is the relationship between N and $\Delta L/L$?

Assumptions

- Each material has the same length and thereby occupies half the total length

Derivation

Assuming each material occupies half the length,

$$L_1 = L_2 = \frac{L}{2}$$

the total elongation is then

$$\Delta L = \epsilon_1 \frac{L}{2} + \epsilon_2 \frac{L}{2}. \quad (14)$$

Substituting the strains found in [Equations \(12\) and \(13\)](#) from Problem 2.1 into [Equation \(14\)](#) we get

$$\Delta L = \frac{N}{a^2 E_1} \frac{L}{2} + \frac{N}{a^2 E_2} \frac{L}{2}.$$

Dividing by L , we get the total elongation

$$\frac{\Delta L}{L} = \frac{N}{2a^2} \left(\frac{1}{E_1} + \frac{1}{E_2} \right).$$

Or, solved for N , as

$$N = \frac{2a^2 E_1 E_2}{E_1 + E_2} \frac{\Delta L}{L}.$$

$$\boxed{\frac{\Delta L}{L} = \frac{N}{2a^2} \left(\frac{1}{E_1} + \frac{1}{E_2} \right)}$$

RESULT

Problem 2.3

What happens if $E_1/E_2 \rightarrow \infty$?

Derivation

From Problem 2.2 we have the elongation as

$$\frac{\Delta L}{L} = \frac{N}{2a^2} \left(\frac{1}{E_1} + \frac{1}{E_2} \right).$$

If we not let $E_1/E_2 \rightarrow \infty$, then

$$\frac{\Delta L}{L} \approx \frac{N}{2a^2 E_2} \implies N \approx a^2 E_2 \frac{\Delta L}{L}.$$

Corresponding to the effective stiffness going towards $E_{\text{eff}} \rightarrow 2E_2$.

In this case, the stiff material barely stretches

$$\epsilon_1 = \frac{N}{a^2 E_1} \rightarrow 0$$

while the soft material still stretches

$$\epsilon_2 = \frac{N}{a^2 E_2}.$$

$$\boxed{\frac{\Delta L}{L} \approx \frac{N}{2a^2 E_2}}$$

RESULT

Problem 2.4

Compare this problem with Problem 1.

Derivation

For Problem 1 we had a composite bar consisting of two materials in parallel and in Problem 2 the materials were in series.

In Problem 1, the perfect bond between the materials meant their elongations must always be equal thereby restricting the elongation of the softer material to that of the stiffer one. In this case, the strains were continuous across the interface, whilst the stresses were not.

In Problem 2, the same force N must pass through both materials in the series configuration, thereby forcing their stresses to be equal. This however leads to unequal elongations, where the softer material is more free to deform.

In other words, the stiffer material dominates for Problem 1 and the softer dominates for Problem 2.

Problem 1:	same strain, different stress
Problem 2:	same stress, different strain.

RESULT